Naman Dhanopiya

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SUMMARY

Computer Science student specializing in Data Science, with hands-on experience in Machine Learning, Android, and Blockchain projects. Built real-world solutions such as chatbots, recommendation systems, and gesture-controlled applications that showcase both technical expertise and practical impact. A strong problem-solver with a solid coding foundation and a passion for creating innovative, user-focused technology, I aim to apply ML and emerging technologies to build impactful solutions that shape the future of intelligent systems.

SKILLS

Languages: Python, C, C++, JAVA, Kotlin, R, SQL
Tools/Platforms: R Studio, Tableau, Advanced Excel, Data analysis, Data mining, Data visualization, Predictive analysis, Git
Framework: Android Studio, DBMS, scikit-learn, TensorFlow, NLTK, OpenCV, Pandas, NumPy
Soft Skills: Problem-Solving Skills, Project Management, Adaptability, Critical Thinking, Teamwork and Collaboration

PROJECTS

CustomSearchEngine [link] — Jun 2025 – Jun 2025

- Created a C++ command-line search engine that finds relevant lines from multiple text files using keywords and fuzzy search.
- Used inverted indexing for fast lookup, TF-IDF for ranking, and built a custom fuzzy matching algorithm using Levenshtein distance.

Outcome: Tested on 25+ text files, retrieving relevant lines in under 0.5 seconds per query with ~97% accuracy in fuzzy word correction and ranking.
Tech stacks used: *C++, Standard Template Library, File Handling, Unordered Maps/Vectors/Sets, TF-IDF, Levenshtein Distance, Command-Line Interface.*

Movie Recommendation System [link] — May 2025 – Jun 2025

- Constructed a personalized movie recommendation system using collaborative filtering to give users personalized suggestions.
- Preprocessed and analyzed user rating data using tidyverse, and developed rating- and similarity-based recommendation models with recommenderlab.

Outcome: Achieved precision of ~0.84, recall of ~0.79, F1-score of ~0.81, and consistent RMSE across folds, indicating stable prediction accuracy.
Tech stacks used: *R, tidyverse, recommenderlab*

TouchlessDocs: Gesture-Operated Document Viewer [link] — Mar 2025 – Apr 2025

- Built a touchless PDF viewer that lets users navigate documents using hand gestures and voice commands, suitable for field professionals.
- Used MediaPipe and OpenCV for gesture recognition, NLTK for voice commands, PyMuPDF for PDF display, and Pickle/CSV for logging and analytics.

Outcome: Tested on 10+ PDFs with 86% gesture recognition accuracy and 90% voice command recognition, improving hands-free document use.
Tech stacks used: *Python, OpenCV, MediaPipe, PyMuPDF, SpeechRecognition, NLTK, Matplotlib, Pandas, Pickle, CSV*

Speech-to-Text Transcription System [link] — Nov 2024 – Dec 2024

- Created a speech-to-text system that converts live or recorded audio into text using advanced models like Wav2Vec2, Whisper, and a custom model.
- Built audio preprocessing pipelines with Librosa, enabled model selection via YAML, and integrated Sounddevice for audio capture and transcription.

Outcome: Successfully transcribed 20+ minutes of audio in multiple sessions with ~92% accuracy, suitable for voice-controlled apps and assistive AI tools.
Tech stacks used: *Python, PyTorch, Transformers, Librosa, Sounddevice, NumPy, ConfigParser/YAML*

TRAINING

GeekForGeeks — May 2024 – Jul 2024

- **In-depth Understanding of Data Structures and Algorithms (DSA):** Developed strong foundation in DSA by learning core concepts such as arrays, linked lists. Progressed to advanced topics including dynamic programming, graph algorithms.
- **Hands-on Problem Solving with Real-World Relevance:** Consistently practiced and solved a wide range of problems from company-specific coding sets, mock tests, and competitive programming platforms.
- **Multi-language Proficiency with Industry-relevant Tech Stacks:** Gained hands-on experience in implementing DSA concepts using multiple programming languages including Python, C++, and Java.

CERTIFICATES

- **Dynamic Programming, Greedy Algorithms** – University of Colorado — **Apr 2024**
- **ChatGPT Advanced Data Analysis** – Vanderbilt University — **Jan 2024**
- **Built AI Apps with ChatGPT, Dall-E, and GPT-4** – Scrimba — **Jan 2024**
- **Generative AI with Large Language Models** – AWS and DeepLearning.AI — **Jan 2024**
- **Prompt Engineering for ChatGPT** – Vanderbilt University — **Jan 2024**

EDUCATION

Lovely Professional University — **Aug 2022 - Present**
Bachelor of Technology - Computer Science and Engineering — Phagwara, Punjab

Samrat Public School — **Apr 2020 – Mar 2021**
Intermediate; Percentage: 93% — Ajmer, Rajasthan

St. Anselm School — **Apr 2018 – Mar 2019**
Matriculation; Percentage: 94% — Ajmer, Rajasthan